

OSBORNE HEAD, NS, Canada

WMO: 713240

Lat: 44.6133N

Lon: 63.4222W

Elev: 30

StdP: 100.97

Time Zone: -4.00 (NAA)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.9	-12.7	-20.2	0.6	-13.3	-18.5	0.7	-11.1	20.9	0.7	18.5	0.7	6.6	280	0.746

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.3	23.6	17.5	21.9	17.5	20.6	17.5	20.0	21.3	19.1	20.3	18.4	19.6	5.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.3	20.6	18.7	13.5	19.7	18.0	13.0	18.9	57.5	21.3	54.4	20.4	52.3	19.6	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.0	12.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-3.3	-3.4	-0.6	3.9	8.4	12.6	16.6	17.9	15.5	10.8	5.3	0.2		
	DBStd	8.21	5.15	4.29	3.91	2.86	2.66	2.95	2.10	1.86	2.68	3.41	3.86	4.80		
	HDD10.0	1858	413	374	330	184	66	8	0	0	1	32	146	303		
	HDD18.3	4150	671	608	587	433	309	172	61	29	91	235	391	562		
	CDD10.0	781	0	0	1	1	15	87	205	245	165	56	5	0		
	CDD18.3	31	0	0	0	0	0	1	8	17	5	1	0	0		
	CDH23.3	68	0	0	0	0	1	12	16	29	8	2	0	0		
	CDH26.7	8	0	0	0	0	0	2	1	5	1	0	0	0		
(14) Wind	WSAvg	5.9	6.8	6.6	6.7	6.1	5.5	5.0	4.6	4.7	5.3	6.2	6.5	6.7		
(15) Precipitation	PrecAvg	1414	132	114	123	110	111	103	101	96	97	128	139	151		
	PrecMax	1795	271	229	243	208	214	349	229	320	225	256	258	253		
	PrecMin	983	42	34	45	19	26	32	13	25	27	32	38	67		
	PrecStd	198	47	47	46	42	45	55	48	61	46	52	55	51		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	6.7	14.2	15.6	21.2	24.7	25.3	26.2	24.4	21.0	15.0	10.6		
		MCWB	7.6	5.9	9.4	9.2	12.6	16.7	17.6	18.8	19.5	15.8	13.4	10.0		
	2%	DB	7.1	5.0	7.9	12.0	16.1	21.4	23.0	23.5	21.7	17.9	13.4	9.5		
		MCWB	6.4	4.3	4.2	6.4	10.3	14.9	17.5	17.7	18.2	16.0	12.2	9.0		
	5%	DB	5.7	3.7	5.7	9.2	14.0	18.8	21.0	21.7	20.2	16.7	12.3	8.4		
		MCWB	5.0	3.0	3.7	5.7	9.5	14.3	17.1	18.4	17.5	15.2	11.4	7.9		
	10%	DB	4.1	2.5	4.2	7.7	12.1	16.6	19.5	20.6	19.3	15.6	11.1	7.3		
		MCWB	3.4	1.8	2.6	5.6	9.2	13.7	17.2	18.2	17.1	14.2	10.0	6.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	6.2	10.7	10.9	14.4	18.6	20.0	21.4	21.4	18.1	14.6	10.2		
		MCDB	8.4	6.3	13.3	14.5	17.9	23.4	22.1	24.2	23.0	19.0	14.9	10.4		
	2%	WB	6.7	4.5	5.6	8.2	12.0	16.5	18.8	20.3	19.7	16.8	12.6	9.1		
		MCDB	6.9	4.9	7.0	9.7	14.1	19.4	20.4	21.6	20.5	17.5	13.0	9.4		
	5%	WB	5.1	3.2	4.1	7.0	10.6	15.2	18.2	19.4	18.7	15.6	11.6	8.0		
		MCDB	5.4	3.5	5.2	8.1	12.7	17.5	19.5	20.5	19.5	16.3	12.2	8.4		
	10%	WB	3.5	1.9	2.8	5.8	9.5	14.1	17.6	18.7	17.7	14.5	10.3	6.8		
		MCDB	4.2	2.4	4.0	7.3	11.3	15.8	18.9	19.7	18.7	15.5	10.9	7.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	6.6	6.5	6.0	6.0	5.9	5.1	5.3	5.9	6.0	5.9	6.6		
		MCDBR	8.2	6.9	8.3	9.4	9.9	10.4	8.5	7.3	6.8	6.4	6.8	7.3		
	5% WB	MCWBR	8.6	7.0	6.9	6.1	5.9	5.6	4.5	3.9	4.6	5.2	6.8	7.7		
		MCWBR	7.9	7.2	7.6	7.6	8.6	8.3	6.0	5.1	5.1	5.6	6.5	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.298	0.324	0.349	0.369	0.400	0.415	0.396	0.353	0.333	0.317	0.297		
		taud	2.474	2.517	2.475	2.448	2.438	2.381	2.359	2.411	2.502	2.554	2.550	2.535		
	Ebn at Noon		821	901	915	914	901	870	851	857	872	846	802	797		
		Edh at Noon	70	80	95	106	111	118	119	109	92	77	65	60		
(44) All-Sky Solar Radiation	RadAvg		1.43	2.33	3.53	4.31	4.95	5.38	5.33	4.99	4.07	2.64	1.52	1.13		
	RadStd		0.11	0.17	0.24	0.35	0.64	0.50	0.39	0.35	0.27	0.27	0.16	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	+0.37	+0.50	N/A	N/A	+56	+17

Nomenclature: See separate page